

Model Analysis Report

Hydrograph Model

MKM-HG-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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# 1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Isolated storm: peak WSE sensitivity ( $d = 636.0\text{m}$ ,  $\Delta z = 1.42\text{m}$ ,  $D = 36\text{h}$ ).

| Peak WSE (m) | Depth (m) | Duration (hrs) | Arrival (hr) | Damage (%) |
|--------------|-----------|----------------|--------------|------------|
| 6.00         | 0.0000    | 0              | —            | 0.0000     |
| 7.00         | 0.2752    | 5              | 12           | 15.0       |
| 8.00         | 1.27      | 10             | 11           | 45.4       |
| 9.00         | 2.26      | 12             | 11           | 64.0       |
| 10.0         | 3.26      | 15             | 11           | 77.6       |
| 12.0         | 5.26      | 18             | 11           | 96.3       |
| 14.0         | 7.27      | 21             | 11           | 100.0      |

Table 2: Doublet storm: peak WSE sensitivity ( $d = 636.0\text{m}$ ,  $\Delta z = 1.42\text{m}$ ,  $D = 36\text{h}$  per pulse).

| Peak WSE (m) | Depth (m) | Duration (hrs) | Arrival (hr) | Damage (%) |
|--------------|-----------|----------------|--------------|------------|
| 6.00         | 0.0000    | 0              | —            | 0.0000     |
| 7.00         | 0.9022    | 8              | 65           | 37.1       |
| 8.00         | 2.05      | 12             | 65           | 60.8       |
| 9.00         | 3.20      | 20             | 12           | 77.0       |
| 10.0         | 4.38      | 27             | 11           | 88.8       |
| 12.0         | 6.69      | 34             | 11           | 100.0      |
| 14.0         | 9.00      | 39             | 11           | 100.0      |

Table 3: Isolated storm: distance sensitivity (peak= 10.0m,  $\Delta z = 1.42\text{m}$ ).

| Distance (m) | Retention | Travel (hrs) | Depth (m) | Duration (hrs) | Damage (%) |
|--------------|-----------|--------------|-----------|----------------|------------|
| 100          | 1.00      | 0.0000       | 3.26      | 15             | 77.6       |
| 250          | 1.00      | 0.0200       | 3.26      | 15             | 77.6       |
| 500          | 1.00      | 0.0500       | 3.26      | 15             | 77.6       |
| 1000         | 1.00      | 0.1300       | 3.26      | 15             | 77.6       |
| 2000         | 0.8187    | 0.3900       | 2.36      | 13             | 65.4       |
| 3000         | 0.7408    | 0.6700       | 1.97      | 12             | 59.4       |
| 5000         | 0.6065    | 1.47         | 1.30      | 11             | 46.0       |

Table 4: Doublet storm: distance sensitivity (peak= 10.0m,  $\Delta z = 1.42\text{m}$ ).

| Distance (m) | Retention | Travel (hrs) | Depth (m) | Duration (hrs) | Damage (%) |
|--------------|-----------|--------------|-----------|----------------|------------|
| 100          | 1.00      | 0.0000       | 4.38      | 27             | 88.8       |
| 250          | 1.00      | 0.0200       | 4.38      | 27             | 88.8       |
| 500          | 1.00      | 0.0500       | 4.38      | 27             | 88.8       |
| 1000         | 1.00      | 0.1300       | 4.38      | 27             | 88.8       |
| 2000         | 0.8187    | 0.3900       | 3.26      | 23             | 77.6       |
| 3000         | 0.7408    | 0.6700       | 2.78      | 21             | 71.7       |
| 5000         | 0.6065    | 1.47         | 1.97      | 17             | 59.4       |

Table 5: Isolated storm: elevation sensitivity (peak= 10.0m,  $d = 636.0\text{m}$ ).

| $\Delta z$ (m) | Threshold (m) | Depth (m) | Duration (hrs) | Damage (%) |
|----------------|---------------|-----------|----------------|------------|
| 0.5000         | 0.8000        | 4.17      | 18             | 86.7       |
| 1.00           | 1.30          | 3.68      | 16             | 81.8       |
| 1.50           | 1.80          | 3.18      | 14             | 76.8       |
| 2.00           | 2.30          | 2.68      | 13             | 70.2       |
| 3.00           | 3.30          | 1.69      | 9              | 53.7       |
| 4.00           | 4.30          | 0.6905    | 5              | 30.7       |
| 5.00           | 5.30          | 0.0000    | 0              | 0.0000     |

Table 6: Doublet storm: elevation sensitivity (peak= 10.0m,  $d = 636.0\text{m}$ ).

| $\Delta z$ (m) | Threshold (m) | Depth (m) | Duration (hrs) | Damage (%) |
|----------------|---------------|-----------|----------------|------------|
| 0.5000         | 0.8000        | 5.30      | 34             | 96.5       |
| 1.00           | 1.30          | 4.80      | 29             | 93.0       |
| 1.50           | 1.80          | 4.30      | 25             | 88.0       |
| 2.00           | 2.30          | 3.78      | 22             | 82.8       |
| 3.00           | 3.30          | 2.78      | 12             | 71.7       |
| 4.00           | 4.30          | 1.79      | 9              | 55.8       |
| 5.00           | 5.30          | 0.7928    | 5              | 33.8       |

Table 7: Isolated storm: pulse duration sensitivity (peak= 10.0m,  $d = 636.0\text{m}$ ,  $\Delta z = 1.42\text{m}$ ).

| Duration (hrs) | Depth (m) | Flooded (hrs) | Arrival (hr) | Peak (hr) | Damage (%) |
|----------------|-----------|---------------|--------------|-----------|------------|
| 12             | 3.26      | 5             | 11           | 11        | 77.6       |
| 24             | 3.26      | 10            | 11           | 12        | 77.6       |
| 36             | 3.26      | 15            | 11           | 13        | 77.6       |
| 48             | 3.26      | 20            | 11           | 14        | 77.6       |
| 72             | 3.27      | 30            | 11           | 16        | 77.7       |
| 96             | 3.28      | 40            | 11           | 19        | 77.8       |
| 120            | 3.28      | 50            | 11           | 21        | 77.8       |

Table 8: Doublet storm: pulse duration sensitivity (peak= 10.0m,  $d = 636.0\text{m}$ ,  $\Delta z = 1.42\text{m}$ ).

| Duration (hrs) | Depth (m) | Flooded (hrs) | Arrival (hr) | Peak (hr) | Damage (%) |
|----------------|-----------|---------------|--------------|-----------|------------|
| 12             | 4.35      | 8             | 11           | 35        | 88.5       |
| 24             | 4.36      | 17            | 11           | 48        | 88.6       |
| 36             | 4.38      | 27            | 11           | 67        | 88.8       |
| 48             | 4.38      | 35            | 11           | 86        | 88.8       |
| 72             | 4.38      | 53            | 12           | 124       | 88.8       |
| 96             | 4.38      | 39            | 12           | 163       | 88.8       |
| 120            | 1.27      | 32            | 12           | 21        | 45.5       |

Table 9: Isolated vs doublet comparison ( $d = 636.0\text{m}$ ,  $\Delta z = 1.42\text{m}$ ,  $D = 36\text{h}$ ).

| Peak (m) | Iso depth (m) | Dbl depth (m) | $\Delta$ depth (m) | Iso dur (hrs) | Dbl dur (hrs) | $\Delta$ dur (hrs) |
|----------|---------------|---------------|--------------------|---------------|---------------|--------------------|
| 6.00     | 0.0000        | 0.0000        | 0.0000             | 0             | 0             | 0                  |
| 7.00     | 0.2752        | 0.9022        | 0.6270             | 5             | 8             | 3                  |
| 8.00     | 1.27          | 2.05          | 0.7838             | 10            | 12            | 2                  |
| 9.00     | 2.26          | 3.20          | 0.9406             | 12            | 20            | 8                  |
| 10.0     | 3.26          | 4.38          | 1.12               | 15            | 27            | 12                 |
| 12.0     | 5.26          | 6.69          | 1.43               | 18            | 34            | 16                 |
| 14.0     | 7.27          | 9.00          | 1.73               | 21            | 39            | 18                 |

## 2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

### 2.1 Stress Scenarios

Table 10: Stress testing scenarios for Hydrograph Model.

| Scenario        | Parameter Shock         | Severity | Status  |
|-----------------|-------------------------|----------|---------|
| Baseline        | No shock                | —        | Passed  |
| Moderate stress | +2 $\sigma$ inputs      | Medium   | Pending |
| Severe stress   | +3 $\sigma$ inputs      | High     | Pending |
| Reverse stress  | Output threshold breach | Critical | Pending |

*Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.*

## 3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 11: Performance metrics for Hydrograph Model.

| <b>Metric</b>               | <b>Value</b>     | <b>Status</b> |
|-----------------------------|------------------|---------------|
| Unit test pass rate         | See Test Results | —             |
| Execution time (single run) | TBD              | —             |
| Numerical stability         | TBD              | —             |
| Reproducibility             | Deterministic    | OK            |

*This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.*